

RAHMA MANNAI

Embedded Software Engineer

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Education

National Engineering School Of Tunis -ENIT-

2020 - 2023

Electrical engineering specializing in Embedded Systems

Preparatory Institute for Engineering Studies of Nabeul -IPEIN-

2018 - 2020

Math-Physics

Work Experience

Embedded Software Engineer

Since September 2023

Focus Corporation Tunisia

- **Embedded Software Engineer on behalf of Continental Automotive Rambouillet**

Final Customer: PSA Group

Team: Customer Diagnosis and TSW

Domain: Connected Vehicle Telematics Unit

- Work according to "Scrum Agile" methodology: Conduct PI and Sprint Planning in Customer Diagnostics Scrum Team.
- Software implementation in compliance with MISRA rules and AUTOSAR specifications.
- Implement Diagnosis Services Routing.
- Configuration and integration of software components with EBTresos from "Elektrobit" according to UDS specifications.
- Experience with diagnostics, telecoding and calibration.
- Bug analysis, bug fix, technical meetings and risk identification.
- Review customer specifications and requirements (IBM-DOORS).
- Experience using Git, Gerrit, Jira and Confluence on Scaled Agile Framework (SAFe) projects.
- Knowledge of continuous integration with Jenkins.
- Component and unit testing.

Keywords: C, C++, Autosar, BSRF, Cortex ARM A7, Cortex ARM M4, SoC, Vector CANAnalyzer/CANoe, EBTresos, DltViewer, Makefiles, Jenkins, XML, Embedded Linux, IBM DOORS..

Projects

End of Studies Project

Feb 2023 - Jul 2023

Embedded software engineer Intern at Focus Corporation Tunisia

- **Authoring tool for AUTOSAR Platform Architecture** The goal of this project is to design and implement an authoring tool for AUTOSAR Platform Architecture. This tool aims to replace the current generator with an innovative interface, this interface streamlines the process of adding new Diagnostic Trouble Codes (DTCs) by automatically integrating them into all relevant output files, ensuring comprehensive system coherence.

Tasks Accomplished:

- Study the workflow of the AUTOSAR Platform Architecture.
- Implementation of the DTCs configuration based on UDS protocol.
- Development of an interface that automates the process of adding new DTCs based on AUTOSAR specifications.
- Test and validate the functionality of the new process.

Keywords: AUTOSAR, UDS protocol, C++, C, EBTresos, ARXML..

Technical Skills

Programming Languages: Embedded C, C++, Python, VHDL.

Tools and Software: EB Tresos, DLT, IBM DOORS, XML, Jenkins.

Process Tools: Git, Guerrit, Gitlab, JIRA, Confluence.

CAN tools: Vector CAN tools (CANFD, CANXL), Vector CANalyser/CANoe.

Architectures: AUTOSAR(AUTomotive Open System ARchitecture), DEM , DCM, BSW.

Operating Systems: Windows, Linux.

Methodology: Scrum

Languages

Arabic: Native language

French : DELF B2

English: TOEIC B2